OUR TEAM

SUNIL MAURYA

Tech & Design

MAHESH GONDA

Development

MEHUL JAIN

Development

SANKET KAMBLE

Product Design

Invent for Customer : YOKOGAWA

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Problem Statement

1. Heavy industry manufacturers currently lack efficient and user-friendly tools to analyse sensor data from their equipment, hindering their ability to optimize production, prevent downtime, and make informed maintenance decisions.
   Sensors type: Temperature, Pressure, Vibration, Energy Meter

2. Lack of a comprehensive dashboard that can effectively correlate electricity consumption data from multiple sources with CO2 emissions

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EcoPulse

Integrated Sensor Monitoring for Sustainable Operations

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Features of EcoPulse

- A **user-friendly** bundle of applications to analyze sensor data which helps in preventing downtimes and make informed maintenance decisions.
- A unified tool to track electricity usage across various sources and its **impact on total carbon footprint**

- **Data Integration**: All sensors will transmit data to the Yokogawa system, and Yokogawa will subsequently send the data to the app. The method of transmission, whether through a file or directly via API, is still under consideration.
- **Sensor and Equipment Mapping**: The dashboard should provide a solution to view and manage the sensors mapped to an equipment
- **Dashboard**: The dashboard should display the health and current data of a given equipment at a specific point in time
- **Business Rules**: Certain conditions may trigger notifications for service based on predefined value ranges or specific situations, leading to planned maintenance orders.
- **Special Sensor:** Certain sensors can send multiple types of data which can be handled within our solution

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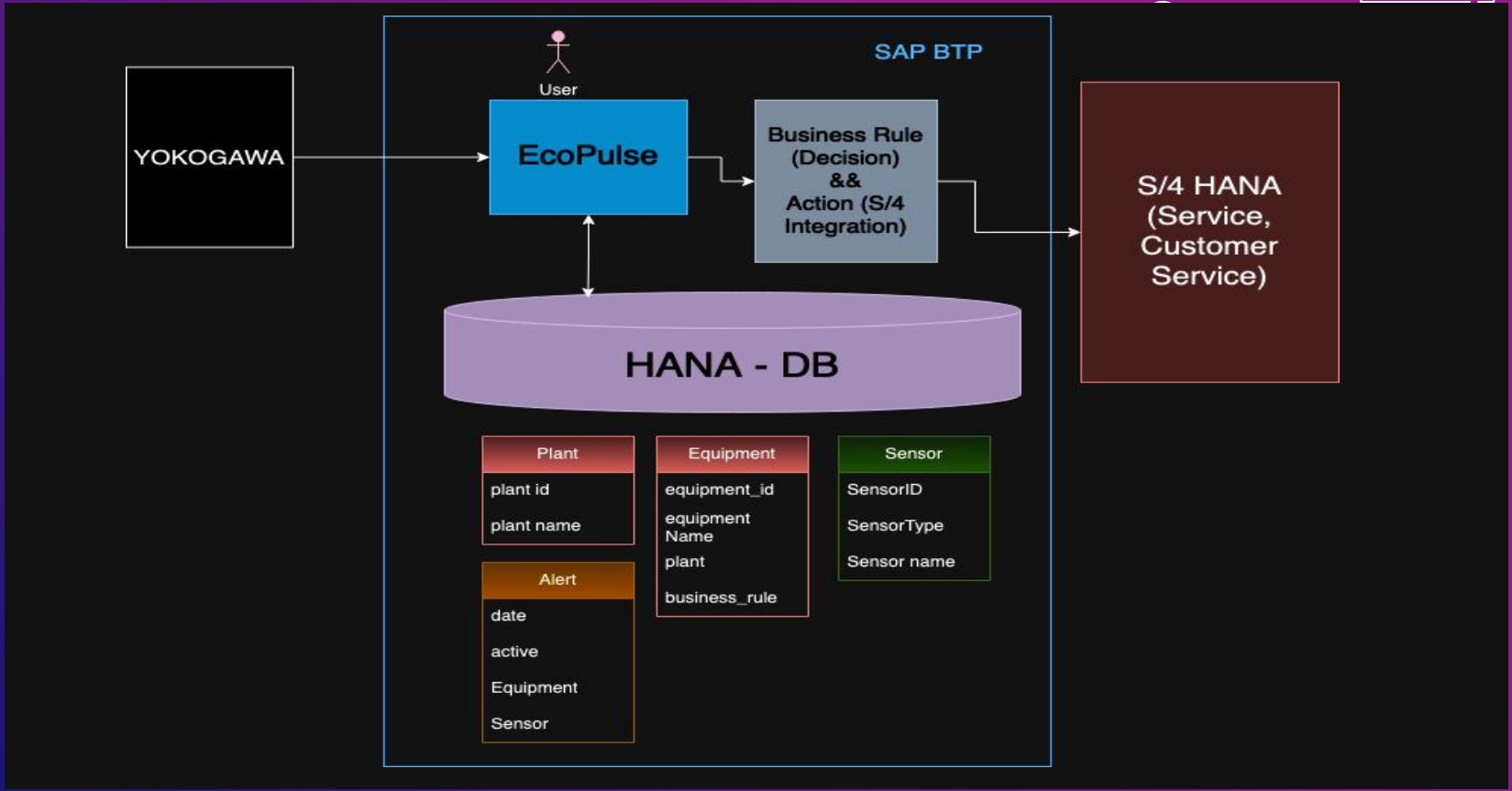
Architecture



PERSONA

Sustainability Manager

- Analyze the carbon footprint
- Take decision/steps to reduce the carbon emissions

Plant Manager

- Analyze the working of all the equipment in the plant and take informed decision about their maintenance

Admin

- Configure the calculations for CO2 emission in line with the local rules and regulations.
- Work with Plant Manager to maintain the configuration master.
- Maintain business rules if required

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UI

Screen flow for Dashboard

Screen flow for Admin

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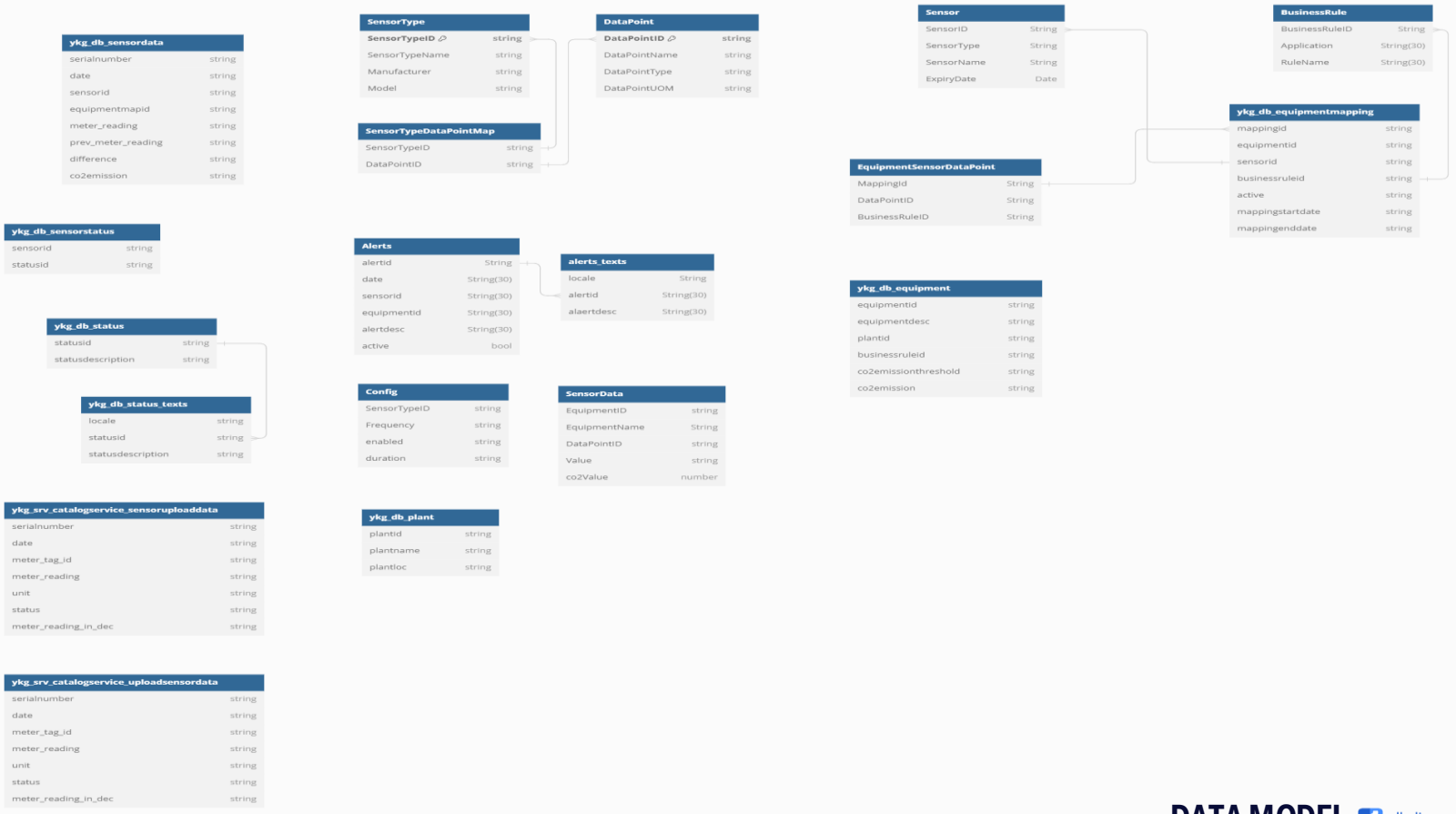
Data Model: For Sensor and Equipments



Solution Demo

Thank You !

Q n A !